SWASTIK CHOWBAY

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Nationality: Indian • Languages: English, Italian (basic)

RESEARCH INTERESTS

High-contrast imaging of protoplanetary disks and embedded protoplanets (VLT/SPHERE star-hopping, JWST); spectroscopic characterization of exoplanet host stars (metallicities, ages and kinematics); Galactic chemical evolution as a driver of planet demographics; and first optical–NIR spectroscopic baselines of newly imaged accreting protoplanets.

PROFESSIONAL EXPERIENCE

Postdoctoral Fellow July 2024 – Present
Università degli Studi di Milano, Milan, Italy

Postdoctoral Researcher April 2024 – June 2024
Indian Institute of Astrophysics, Bengaluru, India

Intern January 2021 – April 2021
European Southern Observatory, Santiago, Chile

EDUCATION

Ph.D. in Physics July 2018 – September 2024
Pondicherry University, with research at the Indian Institute of Astrophysics, Bengaluru (thesis submitted March 2024)
Thesis: "Investigating star-planet properties of different exoplanet populations"
Supervisor: Dr. R. K. Banyal

M.Sc. in Physics July 2016 – May 2018
University of Hyderabad, Hyderabad *(CGPA: 8.67/10)*

B.Sc. in Physics July 2013 – May 2016
University of Calcutta, Kolkata *(70.12%)*

TECHNICAL SKILLS

Instruments	VLT/SPHERE-IRDIS (polarimetry and star-hopping RDI), VLT/ERIS-NIX (L' vortex coronagraph), VLT/MUSE integral-field data, archival VLT/NACO
Post-processing	VIP (full-frame and iterative PCA), KLIP, LOCI; ADI, RDI, star-hopping, and SDI; IRDAP polarimetric reduction
Calibration	Injection–recovery throughput and detection limits; 5σ contrast curves and mass limits (species; AMES-Cond, ATMO 2020); Gaia-anchored astrometric calibration and cross-instrument (ERIS–NACO) astrometry
Modeling	RADMC-3D radiative transfer with CMA-ES optimization; aggregate-grain opacities (AggScatVIR); MCMC (emcee)
Spectroscopy	iSpec, MOOG, Turbospectrum/TSFitPy, IRAF/PyRAF; Bayesian stellar parameters (iMCSpec); Gaia DR3 astrometry and kinematics
Computing	Python (primary), IDL, R, Fortran, C/C++, Java; HPC on CINECA Leonardo; CASA (elementary); Linux, macOS

COMPUTING & HPC ALLOCATIONS

- **Leonardo cluster, CINECA (Italy): 100,000 CPU-hours** (IscrC_PRODISK); PI 2026
- **Delphinus / Nova cluster, IIA (India): ∼10,000 CPU-hours**; PI 2018–2024

CONSORTIA & COLLABORATIONS

- **GPI 2.0 collaboration**: Gemini Planet Imager 2.0 surveys of young exoplanets and disks.
- **ALMA2040 collaboration**: science case for a next-generation (sub-)mm interferometer.

OBSERVATIONS & TELESCOPE PROGRAMS

VLT/SPHERE: awarded as PI (ESO Cycle P116, 2025–2026)

- *Revealing The Moth's Sculptors: SPHERE's Star-Hopping Hunt for Planets and Disk Dynamics in HD 61005* (116.29D7).
- *Unveiling Disk Substructures and Planet Formation: Scattered-Light Imaging of DoAr 33, DoAr 28, and AA Tau* (116.299M); yielded the first scattered-light images of DoAr 33 and HD 135344A.

VLT/SPHERE: awarded as Co-PI (2024–2026)

- *Capturing dusty winds in the Orion proplyds* (116.292Y; PI: S. Facchini).
- *Rings, spirals or planets? A first clear view of the HD 141569A system* (113.269Y; PI: Z. Wahhaj).

ALMA: awarded as PI (Cycle 13, 2026)

- *Feeding the Planet Factory: An ACA Kinematic Test of Late Infall in Taurus* (2026.1.00440.S; ACA Band 6, $^{12/13}$CO and C^{18}O J=2−1, 12 SPHERE-selected Taurus targets, 27.1 h total).

JWST: awarded as Co-I (Cycle 5, 2026)

- *The first observation of H_2 absorption in a protoplanetary disk through transmission spectroscopy* (Program 11087; NIRSpec IFU; PI: F. M. Alarcon).

Himalayan Chandra Telescope (HCT): 4 awarded PI programs (2020–2026)

- *Anchoring an Accreting Protoplanet: the First Optical–NIR Spectroscopic Baseline of WISPIT-2* (HCT-2026-C2-P47).
- *Long-term monitoring of the WISPIT-2 host star* (HCT 2026).
- *Determination of host-star metallicity of directly imaged planets* (HCT-2020-C2-P34).
- *Determination of host-star metallicity of directly imaged brown dwarfs* (HCT-2020-C3-P10).

Other observational work

- Awarded ∼100 hours over three cycles on HFOSC/Himalayan Chandra Telescope, Hanle.
- HIP 41378f transit campaign: PI for India sites (GROWTH-India, Hanle; JCBT, Kavalur).

PUBLICATIONS

Summary: 14 refereed publications | 8 as first author (7 in astronomy) | 7 submitted or in preparation
Google Scholar: **231** citations | h-index: **7** | i10-index: **7** *(as of August 2026)*

Major research contributions

- **Stellar hosts of directly imaged planets and brown dwarfs:** showed that the hosts of directly imaged wide-orbit giant planets and brown dwarfs have a broadly solar metallicity distribution, without the metal-rich bias of close-in giant-planet hosts, which constrains how wide companions form.
- **Galactic chemical evolution and planet demographics:** showed that host-star abundance and age trends are shaped by Galactic chemical evolution, with giant-planet hosts systematically younger than small-planet hosts in Gaia DR3 chemical and kinematic age proxies and in isochrone ages.
- **Star-hopping high-contrast imaging:** delivered self-subtraction-free polarimetric and total-intensity SPHERE images of the LkCa 15 disk, found that the measured phase function and polarized fraction favor porous aggregates over compact grains in its surface, and recovered three survey disks not reported by the original reduction.
- **New disk detections:** first scattered-light images of DoAr 33 and of a disk around HD 135344A, host of the planet HD 135344 Ab (manuscripts in preparation).

First-author and lead-contributor publications

1. *Imaging the LkCa 15 system in polarimetry and total intensity without self-subtraction artefacts.* **C. Swastik**, Z. Wahhaj, M. Benisty, S. Arora, C. Ginski, B. B. Ren, R. G. van Holstein, R. de Rosa, R. K. Banyal, R. Tazaki. *Astronomy & Astrophysics* **706**, A312 (2026).

2. *A quiet host in an active planet-forming disk: optical spectroscopy of WISPIT 2.* **C. Swastik**, M. Bestha, L. S. Sonith, S. Facchini, T. Sivarani, R. K. Banyal, Z. Wahhaj, A. Choudhary, M. Gopinathan, P. Saraf, A. Mallick, M. P. Navaneeth, S. Biswas, K. Khushbu. *Astronomy & Astrophysics* **712**, L13 (2026).

3. *Host-star metallicities and kinematics of directly imaged brown-dwarf companions.* **C. Swastik**, R. K. Banyal, M. Muduli, S. Soni, A. Jino, S. Facchini, Z. Wahhaj, G. Lodato, P. Saraf, A. Choudhary, Bhavya A. K., M. P. Navaneeth, B. Banerjee, S. Biswas, T. Sivarani, G. Maheswar, A. Surya. *Astronomy & Astrophysics*, accepted for publication (2026).

4. *Age analysis of extrasolar planets: insight from stellar isochrone models.* **C. Swastik**, R. K. Banyal, M. Narang, A. Unni, T. Sivarani. *Astronomical Journal* **167**, 270 (2024).

5. *Age distribution of exoplanet host-stars: chemical and kinematic age proxies from Gaia DR3.* **C. Swastik**, R. K. Banyal, M. Narang, A. Unni, B. Banerjee, P. Manoj, T. Sivarani. *Astronomical Journal* **166**, 91 (2023).

6. *Galactic chemical evolution of exoplanet hosting stars: are high-mass planetary systems young?* **C. Swastik**, R. K. Banyal, M. Narang, P. Manoj, T. Sivarani, S. P. Rajaguru, A. Unni, B. Banerjee. *Astronomical Journal* **164**, 60 (2022).

7. *Host-star metallicity of directly imaged wide-orbit planets: implications for planet formation.* **C. Swastik**, R. K. Banyal, M. Narang, P. Manoj, T. Sivarani, B. E. Reddy, S. P. Rajaguru. *Astronomical Journal* **161**, 114 (2021).

8. *Bouncing Universe with exotic radiation.* **C. Swastik**, P. K. Suresh, B. Maity. *Reports in Advances of Physical Sciences* **4**, 2050001 (2020). *(M.Sc. thesis project; theoretical cosmology.)*

Co-authored publications

9. *EK Cha b: multi-tracer detection of a protoplanet candidate in a gas-rich environment.* F. Alarcón, **C. Swastik**, S. Facchini, E. A. Bergin, G. Cugno. *Astronomy & Astrophysics* (Letter), accepted for publication (2026).

10. *The 10 pc neighborhood of habitable zone exoplanetary systems: threat assessment from stellar encounters & supernovae.* T. Pyne, R. K. Banyal, **C. Swastik**, A. De. *Astronomical Journal* **169**, 13 (2025).

11. *Exoplanet classification through Vision Transformers with temporal image analysis.* A. Choudhary, S. Bandari, B. S. Kushvah, **C. Swastik**. *Astronomical Journal* **170**, 120 (2025).

12. *PDS 70 unveiled by star-hopping: total intensity, polarimetry, and millimeter imaging modeled in concert.* Z. Wahhaj, M. Benisty, C. Ginski, S. Arora, **C. Swastik**, R. G. van Holstein, R. de Rosa, B. Yang, J. Bae, B. Ren. *Astronomy & Astrophysics* **687**, A257 (2024).

13. *Protoplanetary disks in K_s-band total intensity and polarized light.* B. B. Ren, M. Benisty, C. Ginski, R. Tazaki, N. L. Wallack, J. Milli, A. Garufi, J. Bae, S. Facchini, F. Ménard, P. Pinilla, **C. Swastik**, R. Teague, Z. Wahhaj. *Astronomy & Astrophysics* **680**, A114 (2023).

14. *Carbon abundance of stars in the LAMOST–Kepler field.* A. Unni, M. Narang, T. Sivarani, M. Puravankara, R. K. Banyal, A. Surya, S. P. Rajaguru, **C. Swastik**. *Astronomical Journal* **164**, 181 (2022).

MANUSCRIPTS SUBMITTED OR IN PREPARATION

As lead author

1. *ERIS/NIX L'-band vortex coronagraph observations of the transitional disks J1842 and V4046 Sgr.* **C. Swastik**, S. Facchini, D. Fedele, V. Roccatagliata, G. Cugno, V. Christiaens, Exo-ALMA team (Under review in A&A).

2. *Small grains in the most compact single-star DSHARP disc reach the CO radius.* **C. Swastik**, Bhavya A. K., M. P. Navaneeth, R. K. Banyal (To be submitted to A&A Letters).

3. *A controlled comparison of confirmed exoplanet host samples from Kepler, K2, and TESS.* **C. Swastik**, S. Soni (To be submitted in AJ).

4. *A polarised scattered-light disk and an embedded companion around HD 135344 A: SPHERE star-hopping reveals a planet within a debris-dominated environment around the binary primary.* **C. Swastik**, S. Facchini, Z. Wahhaj, R. Banyal, Navaneeth M. P., M. Benisty, et al (To be submitted to ApJL / Nature Astronomy).

5. *Star-Hopping Scattered-Light Observations of the V4046 Sgr Circumbinary Disk.* **C. Swastik**, Z. Wahhaj, R. Banyal, S. Facchini, M. Benisty, et al (Manuscript in prep).

Co-authored

6. *Thorium abundances in FGK planet-host stars: Galactic chemical evolution and a planet-host signature.* A. Choudhary, P. Saraf, **C. Swastik**, B. S. Kushvah, R. K. Banyal, T. Sivarani, M. Gopinath (Under review in ApJ).

7. *Star-hopping scattered-light imaging and combined ALMA–SPHERE modeling of the HD 163296 protoplanetary disk.* Navaneeth M. P., Z. Wahhaj, **C. Swastik**, R. K. Banyal et al. (To be submitted to A&A).

AWARDS & GRANTS

- **ERC Starting Grant UNVEIL** (No. 101076613), postdoc funding (PI S. Facchini). 2024–present
- **ICTS Visiting Scientist support** (ICTS/EXOPLANETS2026/03), *Exploring Exoplanets.* 2026
- **PRIN-MUR (Italy), 2022YP5ACE**; financial contribution. 2024
- **COSPAR Research and Travel Grant, ∼EUR 800**, for attending COSPAR-2024, Busan. 2024
- **DST International Travel Grant, INR 150,000**, *Towards Other Earths III*, Porto. 2023
- **IIA Travel Grant, ∼EUR 3,000**, for international conference travel. 2022
- Best poster award, *Pressing for Progress: Gender Equity in Physics.* 2019
- Qualified NET-JRF (PhD eligibility, India) 2017
- Top 15 of 11,535 in National Graduate Physics Examination 2016

SELECTED CONFERENCES AND MEETINGS

- Discs on the Exe, Univ. of Exeter, UK, *Oral* Jul 2026
- Exploring Exoplanets, ICTS-TIFR, Bengaluru, India, *Oral* Mar 2026
- Spirit of Lyot 6, Caltech, Pasadena, USA, *Poster* Feb 2026
- EXO-ELT 2025, ESO, Garching, Germany, *Poster* Nov 2025
- GRC: Origins of Solar Systems, Mt. Holyoke, MA, USA, *Poster* Jun 2025
- GAP: Gas Accretion in Planet Formation, MPIA, Heidelberg, Germany, *Oral* Mar 2025
- Italian National Conf. on Star & Planet Formation, Sexten, Italy, *Oral* Jan 2025
- Alive Universe, Azerbaijan, *Oral* Sep 2024
- COSPAR-2024, Busan, South Korea, *Oral* Jul 2024
- Exoplanet Conference, IISER Pune, *Oral* Aug 2023
- Towards Other Earths III, Porto, Portugal, *Poster* Jul 2023
- Protostars and Planets VII, Kyoto, Japan, *Poster* Apr 2023
- Planet–ESLAB 2023, ESA, Netherlands, *Poster* Mar 2023
- 41st ASI Meeting, IIT Indore, *Oral* Mar 2023

INVITED TALKS & RESEARCH VISITS (EACH WITH A DEPARTMENTAL SEMINAR)

- **TIFR**, Mumbai, India; host: P. Chaturvedi — Jan 2026
- **S. N. Bose National Centre**, Kolkata, India; host: R. Mondal — Dec 2025
- **URSC / ISRO**, Bengaluru, India; host: S. Narendranath — Sep 2025
- **IIA**, Bengaluru, India; host: R. K. Banyal — Aug 2025
- **IIT (ISM) Dhanbad**, India; host: B. S. Kushvah — Aug 2025
- **MIT**, Cambridge, USA; host: S. Seager — Jul 2025
- **NISER**, Bhubaneswar, India; host: L. Majumdar — Jul 2025
- **IISER Berhampur**, India; host: B. Das — Jul 2025
- **INAF–Arcetri**, Florence, Italy; host: D. Fedele — Feb 2025

STUDENTS MENTORED

- **Catherine John** (M.Sc., Christ University, Bengaluru): *Demographics of exoplanets: analysis of the mass–radius relation.*
- **Aarthi Krishna** (M.Sc., IISER Tirupati): *Understanding the brown-dwarf desert by analysing the host stars of directly imaged brown dwarfs.*
- **Athul Rathnakar** (M.Sc., St. Xavier's University, Bengaluru): *Investigating the stellar ages of FGK stars using stellar isochrone models.*
- **Satyam Soni** (M.Sc., NIT Rourkela): *Investigating the brown-dwarf desert for directly imaged exoplanets.*
- **Tisyagupta Pyne** (M.Sc., Visva-Bharati University, Santiniketan): *The 10 pc stellar neighbourhood of habitable zone planetary systems.*
- **Mritunjaya Muduli** (Visvesvaraya Technological University, Belagavi): *Computational analysis of protoplanetary disks.*
- **Anupma Choudhary** (IIT (ISM) Dhanbad): *Thorium abundances of planet-hosting stars.*

SERVICE & OUTREACH

- **Peer review:** referee for *Astronomical Journal* (AJ) and *Astronomy & Astrophysics* (A&A).
- **Professional membership:** Astronomical Society of India (ASI).
- Student representative on IIA's outreach committee; led sky-watch programs for underprivileged children and teacher-training programs for government school teachers around Bengaluru (IAU-supported).
- Co-author of thematic proposal *New Worlds: Are we alone?*, selected under Open Astronomy Schools, IAU-100; volunteer with animal-welfare organizations.
- **Media coverage:** Research featured by India's Department of Science & Technology and the Ministry of Information & Broadcasting.

REFEREES

- **Dr. Stefano Facchini**, Postdoc Supervisor; Assoc. Prof., UniMi; *stefano.facchini@unimi.it*
- **Dr. Zahed Wahhaj**, Lead Collaborator; Staff Astronomer, ESO; *zwahhaj@eso.org*
- **Dr. Ravinder K. Banyal**, PhD Supervisor; Assoc. Prof., IIA, Bengaluru; *banyal@iiap.res.in*
- **Dr. Sivarani Thirupathi**, Collaborator; Professor, IIA, Bengaluru; *sivarani@iiap.res.in*
- **Dr. Maheswar Gopinathan**, Doctoral Committee; Assoc. Prof., IIA, Bengaluru; *maheswar.g@iiap.res.in*